
Design Document

for

Centralized Mess Management System (CMMS)

Version 1.0

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Aaditya Bijarniya Shubham Kumar Dev Prakash Ashwina Yadav Shital Anurath Niras Kavita Kumari Tanush Khagwal Deepak Kumar Itesh Mishra Soumyaranjan Sahu	First Draft of Design Documentation for the project CMMS.	06/02/26

1 Context Design

1.1 Context Model

The Centralized Mess Management System (CMMS) is a web-based platform designed to digitize and integrate mess operations across multiple hostels into a single centralized system. The primary objective of CMMS is to streamline student dining activities such as menu viewing, extra item booking, rebate application, and expense tracking, while simultaneously providing mess staff and administrators with tools for order validation, inventory management, and grievance handling.

The CMMS acts as a central hub connecting multiple external actors and functional subsystems. It facilitates seamless interaction between students, mess staff, and administrators, while maintaining a secure backend for processing bookings, QR validation, rebates, and financial records.

1.1.1 Actors and Their Roles

Student (Hall Resident):

- Views centralized daily and weekly mess menus
- Books extra food items and receives QR codes
- Applies for mess rebate (leave)
- Tracks monthly expenditure
- Submits feedback or complaints

Mess Staff (Worker):

- Scans QR codes for extras and guest meals
- Verifies booking authenticity
- Updates order delivery status

Mess Administrator (Manager):

- Uploads and updates mess menus
- Manages extra item availability
- Reviews and processes rebate applications
- Handles student complaints
- Monitors bookings and inventory

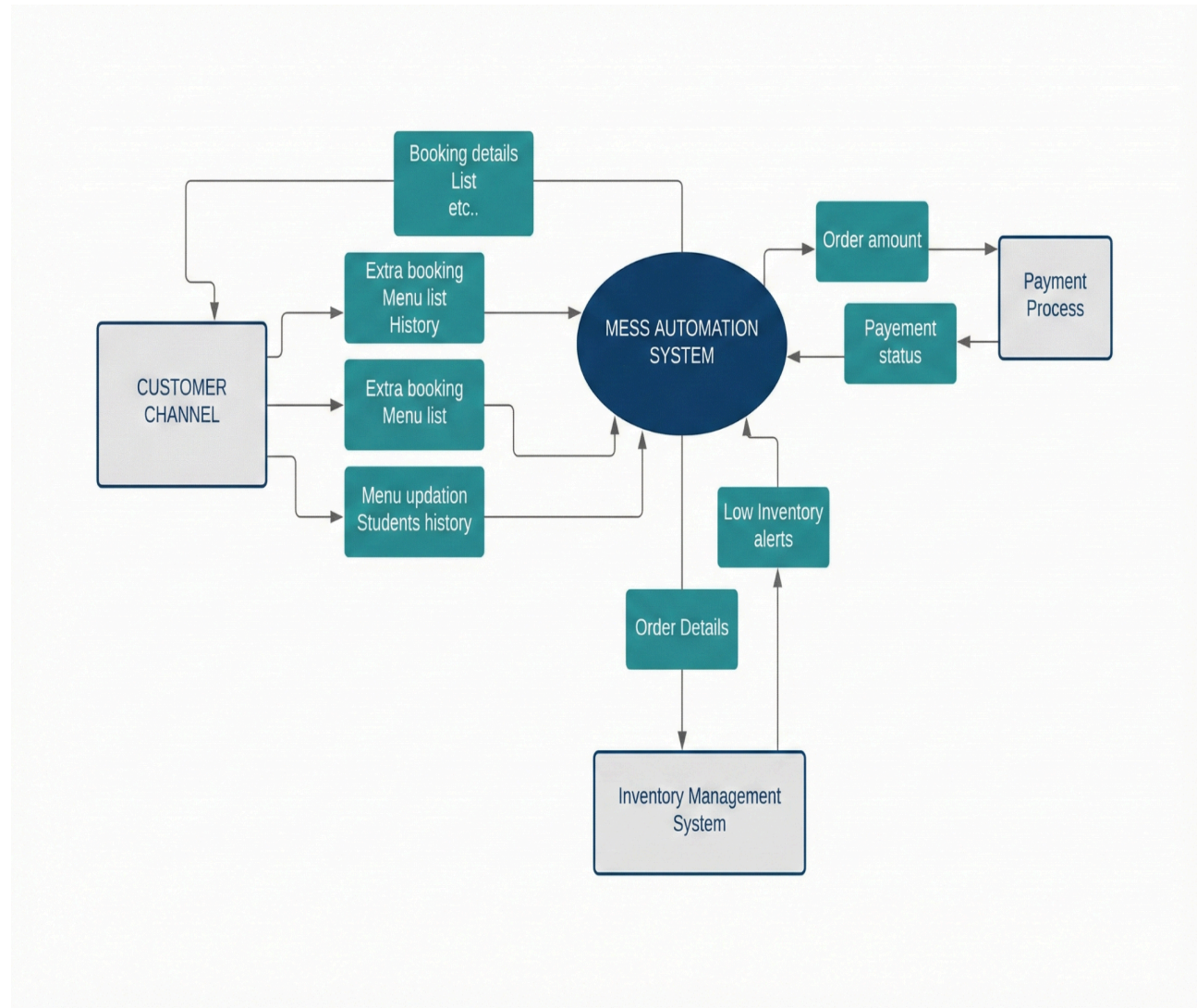
1.1.2 System Components

The CMMS interacts with the following internal subsystems:

- **Authentication Module:** Handles user login, role verification, and secure access control
- **QR Validation Module:** Generates and validates single-use QR codes
- **Rebate Management Module:** Accepts, validates, and processes mess leave applications
- **Inventory Management Module:** Maintains live stock levels and prevents overbooking
- **Expense Tracking Module:** Records student expenditures
- **Feedback Module:** Stores and manages complaints and feedback

All transactional and user data are stored in a centralized database. The frontend interfaces communicate with backend services through secure APIs, ensuring consistency, reliability, and real-time synchronization across all mess operations.

This context model establishes CMMS as the core system coordinating interactions among students, mess staff, administrators, and supporting services, enabling efficient, transparent, and paperless mess management.



1.2 Human Interface Design

The CMMS web application provides a clean, role-based, and intuitive interface for students, mess staff, and administrators. Each screen is designed to minimise user effort while ensuring transparency in mess operations. The major interfaces implemented in the system are described below.

1.2.1 Home / Landing Page

Purpose: Serves as the entry point to the system and allows users to select their role (Student, Mess Staff, or Manager).

Key Elements:

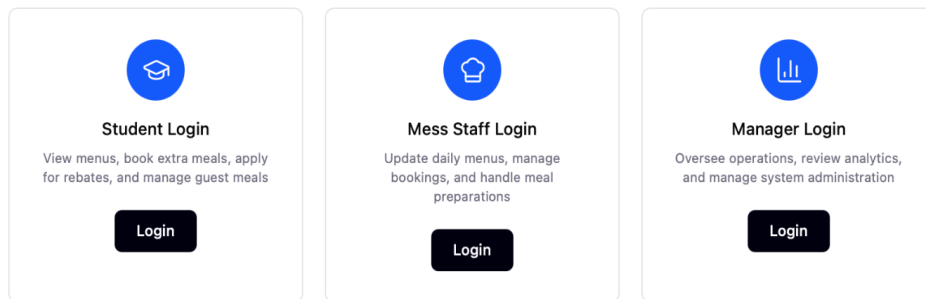
- System title and brief description
- Three role-based login cards
- Navigation header

User Interaction: Users select their role and proceed to the corresponding login page.



Centralized Mess Management System

CMMS digitizes mess operations by streamlining menu planning, extra meal bookings, rebate management, guest meal tracking, and provides comprehensive analytics for efficient cafeteria administration.



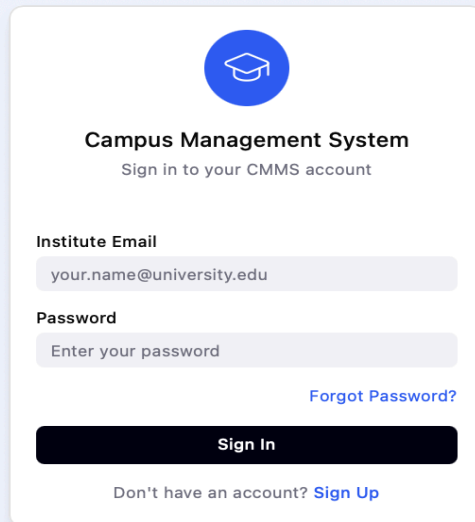
1.2.2 Login Page

Purpose: Authenticates users using institutional email and password.

Key Elements:

- Email input field
- Password input field
- Forgot Password link
- Sign In button
- Sign Up option

User Interaction: User enters credentials, which are verified by the backend before redirecting to the dashboard.



The image shows a login form for the Campus Management System (CMMS). At the top, there is a blue circular icon containing a white graduation cap. Below the icon, the text "Campus Management System" is displayed in bold, followed by "Sign in to your CMMS account" in a smaller font. The form contains two input fields: "Institute Email" with the placeholder text "your.name@university.edu" and "Password" with the placeholder text "Enter your password". To the right of the password field is a blue link labeled "Forgot Password?". Below the input fields is a dark blue button with the text "Sign In" in white. At the bottom of the form, there is a link that says "Don't have an account? Sign Up" in blue.

1.2.3 Mess Menu (Single Day & Weekly View)

Purpose: Allows students to view daily or weekly meal schedules for the selected hostel.

Key Elements:

- Day selector / Weekly toggle
- Breakfast, Lunch, and Dinner sections
- Categorised food items (Main, Curry, Side, Dessert, Beverage)
- Hostel selector

User Interaction: Users can switch between single-day and weekly views to browse upcoming meals.

Mess Menu

Weekly meal schedule

Hostel: Hostel 1

<Monday>

Single DayWeekly View

Breakfast

Idli	Main
Sambar	Curry
Coconut Chutney	Side
Filter Coffee	Beverage

Lunch

Steamed Rice	Main
Chapati	Main
Dal Tadka	Curry
Mixed Vegetable Curry	Curry
Curd	Side
Papad	Side
Pickle	Condiment

Dinner

Chapati	Main
Jeera Rice	Main
Rajma Masala	Curry
Aloo Gobi	Curry
Raita	Side
Gulab Jamun	Dessert

Mess Menu

Weekly meal schedule

Hostel: Hostel 1

Single Day Weekly View

Monday	Tuesday	Wednesday	Thursday	Friday																																																														
Breakfast <table><tr><td>Idli</td><td>Main</td></tr><tr><td>Sambar</td><td>Curry</td></tr><tr><td>Coconut Chutney</td><td>Side</td></tr><tr><td>Filter Coffee</td><td>Beverage</td></tr></table>	Idli	Main	Sambar	Curry	Coconut Chutney	Side	Filter Coffee	Beverage	Breakfast <table><tr><td>Masala Dosa</td><td>Main</td></tr><tr><td>Sambar</td><td>Curry</td></tr><tr><td>Tomato Chutney</td><td>Side</td></tr><tr><td>Tea</td><td>Beverage</td></tr></table>	Masala Dosa	Main	Sambar	Curry	Tomato Chutney	Side	Tea	Beverage	Breakfast <table><tr><td>Upma</td><td>Main</td></tr><tr><td>Vada</td><td>Main</td></tr><tr><td>Coconut Chutney</td><td>Side</td></tr><tr><td>Coffee</td><td>Beverage</td></tr></table>	Upma	Main	Vada	Main	Coconut Chutney	Side	Coffee	Beverage	Breakfast <table><tr><td>Poha</td><td>Main</td></tr><tr><td>Jalebi</td><td>Sweet</td></tr><tr><td>Tea</td><td>Beverage</td></tr></table>	Poha	Main	Jalebi	Sweet	Tea	Beverage	Breakfast <table><tr><td>Pon</td><td></td></tr><tr><td>Sam</td><td></td></tr><tr><td>Coc</td><td></td></tr><tr><td>Cof</td><td></td></tr></table>	Pon		Sam		Coc		Cof																									
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1.2.4 Order Extra Items

Purpose: Enables students to book additional food items and guest meals.

Key Elements:

- Item cards with images, descriptions, prices, and availability status
- Add to Order buttons

User Interaction: Students add items to their cart and confirm bookings, after which a QR code is generated.

Order Extra Items

Select from our menu of additional food items and guest meals



Matar Paneer

₹180

Cottage cheese and green peas in spiced gravy

+ Add to Order



Chicken Biryani

₹250

Aromatic basmati rice with tender chicken pieces

+ Add to Order



Egg Biryani

₹150

Fragrant rice with boiled eggs and spices

+ Add to Order



Papdi Chaat

₹80

Crispy wafers with yogurt, chutneys and spices

+ Add to Order



Gulab Jamun

₹60

Soft milk dumplings in rose-flavored sugar syrup

+ Add to Order



Imarti Rabdi

₹100

Crispy imarti served with creamy thickened milk

+ Add to Order

1.2.5 Mess Bill Summary

Purpose: Displays a consolidated monthly bill including base charges, extras, guest meals, and rebates.

Key Elements:

- Attendance summary
- Base mess charges
- Extra item expenses
- Guest meal charges
- Rebates applied
- Final payable amount

User Interaction: Students review their monthly expenditure and due amount.

Mess Bill Summary		ST2024001 Rahul Kumar
January 2026		
Attendance		28 / 31 days
Base Mess Charges		
Monthly Mess Fee		₹ 4,500
Fixed monthly charge		
Meal Count		₹ 2,800
28 days × ₹100/day		
Subtotal		₹ 7,300
Extra Item Expenses		
Special Dinner		₹ 450
3 occasions		
Tea/Coffee		₹ 280
28 servings		
Extra Snacks		₹ 320
8 items		
Subtotal		₹ 1,050
Guest Meal Charges		
Guest Breakfast		₹ 200
2 guests		
Guest Lunch		₹ 450
3 guests		
Subtotal		₹ 650
Rebates Applied		
Absence Rebate		₹ 300
3 days absence		
Total Rebate		₹ 300
Total Payable Amount		₹ 8,700
Due by 10th February 2026		

1.2.6 Activity History

Purpose: Allows students to track past bookings, guest meals, and rebate history.

Key Elements:

- Tabbed view (Extra Bookings, Guest Bookings, Rebate History)
- Transaction table with date, item, amount, and status

User Interaction: Users monitor completed, pending, or cancelled transactions.

Activity History

View and manage your past bookings and transactions

 Extra Bookings

 Guest Bookings

 Rebate History

DATE	ITEM / TYPE	AMOUNT	STATUS
Feb 5, 2026	Paneer Paratha	₹ 60.00	✓ Completed
Feb 3, 2026	Milk	₹ 25.00	✓ Completed
Feb 1, 2026	Milk	₹ 25.00	⌚ Pending
Jan 28, 2026	Boiled Eggs	₹ 30.00	✓ Completed
Jan 25, 2026	Veg Biryani	₹ 80.00	⊗ Cancelled

Showing 5 records

1.2.7 Student Feedback & Complaints

Purpose: Provides a digital grievance system for reporting issues and tracking resolution status.

Key Elements:

- Complaint category dropdown
- Description textbox

- Submit button
- List of previously submitted complaints with status indicators

User Interaction: Students submit feedback and view updates (Open, In Progress, Resolved).

Student Feedback & Complaints

Submit your feedback or complaint and track its status

Submit New Complaint or Feedback

Category

Select a category



Description

Please describe your complaint or feedback in detail...

 Submit

Previously Submitted Complaints

Hygiene

 In Progress

Poor hygiene conditions in the mess. Tables are not cleaned properly after meals

Submitted on January 30, 2026

Food

 Resolved

Please add more fast food options like burgers and pizzas to the menu

Submitted on January 25, 2026

Food

 Open

Please remove manchurian from the menu as it is not prepared well

Submitted on February 1, 2026

1.2.8 Mess Rebate Application

Purpose: Allows students to apply for mess leave rebates and track application outcomes.

Key Elements:

- Eligibility guidelines
- Leave start and end date selectors
- Apply for Rebate button
- Application status history with approved, processing, and rejected entries

User Interaction: Students submit rebate requests and view credited amounts or rejection remarks.

Student Mess Rebate Application

Apply for mess leave rebates and track your application status

○ **Eligibility & Deadlines:**

- Minimum leave duration: 3 consecutive days
- Applications must be submitted at least 2 days before leave start date
- Rebate rate: ₹150 per day
- Maximum 30 days of leave per semester
- Applications submitted after the deadline will not be processed

New Rebate Application

Select your mess leave dates to apply for a rebate

Leave Start Date

Leave End Date

 **Apply for Rebate**

Application Status History

View the status of your previous rebate applications

RBT-2026-001 Approved Leave Period: Jan 15, 2026 to Jan 20, 2026 Duration: 6 days Submitted: Jan 10, 2026 Remarks: Rebate credited to account	Amount ₹900
RBT-2025-089 Approved Leave Period: Dec 22, 2025 to Dec 28, 2025 Duration: 7 days Submitted: Dec 18, 2025 Remarks: Rebate credited to account	Amount ₹1,050
RBT-2025-078 Processing Leave Period: Nov 10, 2025 to Nov 14, 2025 Duration: 5 days Submitted: Nov 8, 2025 Remarks: Under review by mess committee	Amount ₹750
RBT-2025-056 Rejected Leave Period: Oct 2, 2025 to Oct 3, 2025 Duration: 2 days Submitted: Oct 1, 2025 Remarks: Duration less than minimum 3 days	Amount ₹300

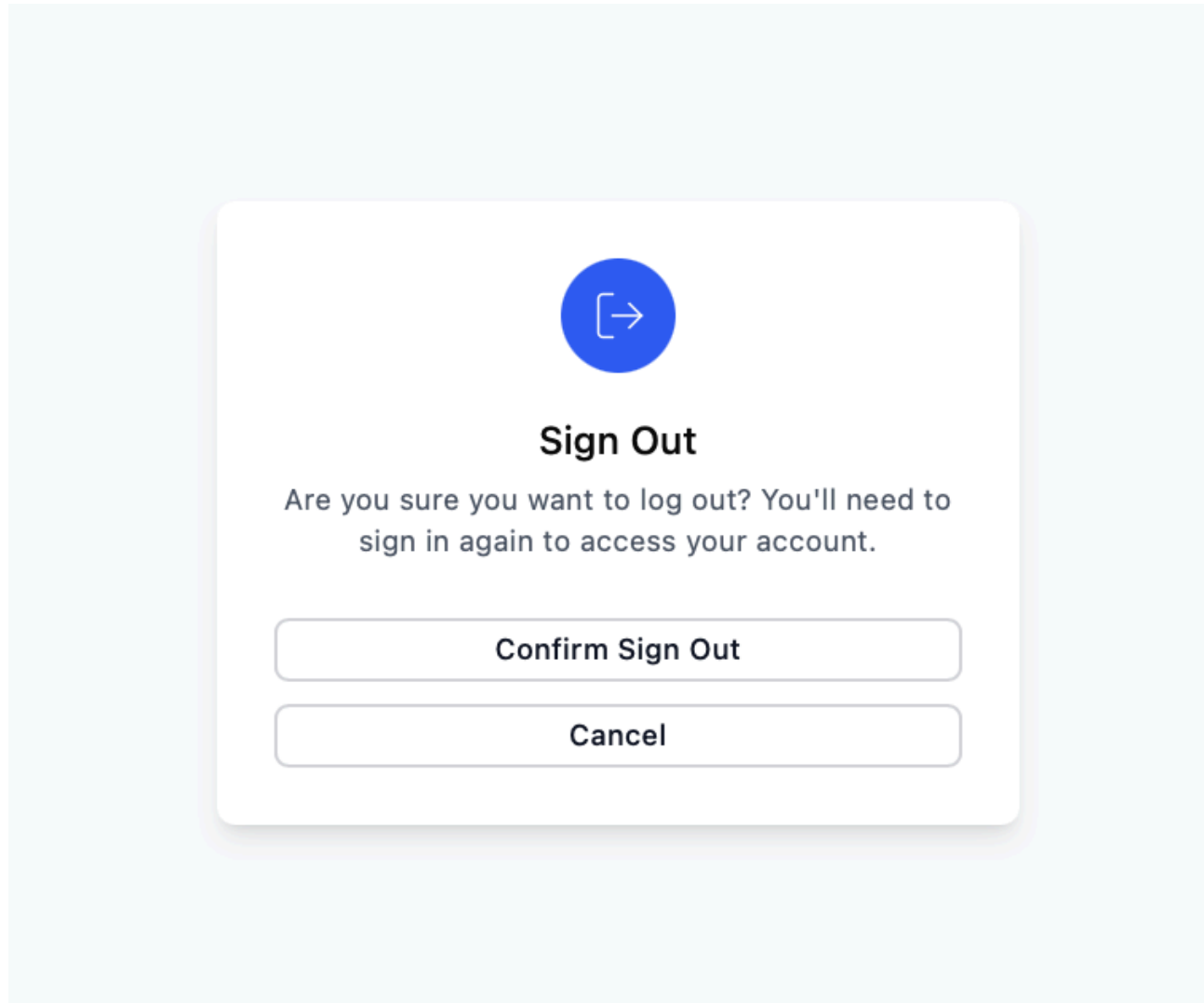
1.2.9 Sign Out

Purpose: Securely logs the user out of the system.

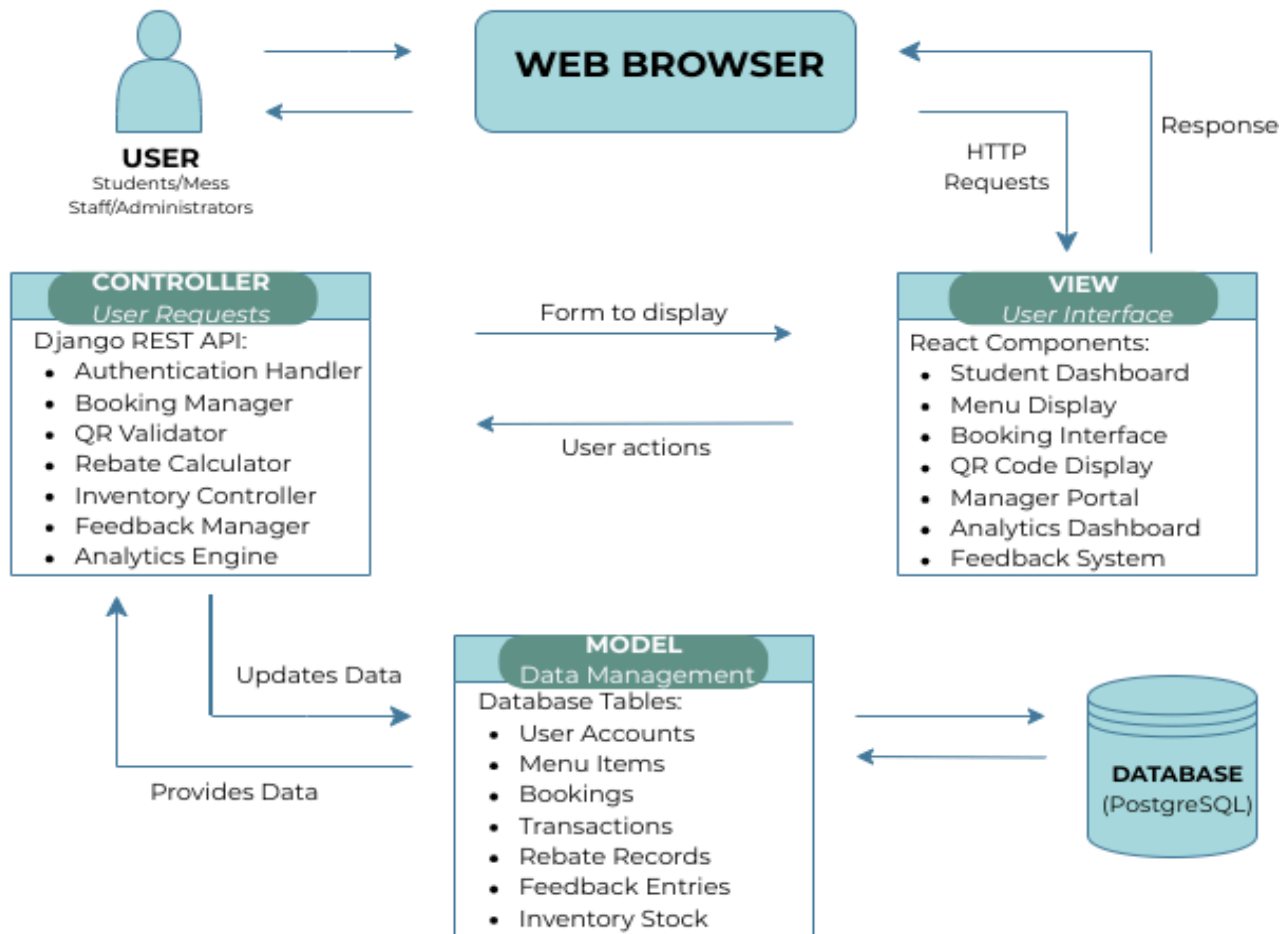
Key Elements:

- Confirmation dialog
- Confirm and Cancel buttons

User Interaction: User confirms logout and is redirected to the landing page.



2 Architecture Design



1.1 Intuition

The MVC architecture is optimal for CMMS as it effectively structures the web-based mess management system into distinct functional layers. The **presentation layer (View)** handles UI elements like the QR-code display for students and the scanner for staff. The **business logic layer (Controller)** manages the heavy lifting, such as calculating automated rebates and preventing "double-bookings" when two students attempt to purchase the last available item simultaneously. Finally, the **data layer (Model)** ensures secure, centralized storage of menus and expenses across all campus hostels. This separation enables modular development, where interface changes can be made without affecting the core inventory and billing logic.

The MVC architectural pattern provides CMMS with a robust, scalable, and maintainable foundation that directly addresses the system's functional and non-functional requirements. The clear separation of presentation, business logic, and data persistence layers enables independent development and deployment while ensuring security, performance, and reliability.

1.2 How the Architecture Meets Non-Functional Requirements

1.2.1 Performance

- Independent layers allow optimization without cross-impact.
- REST APIs ensure fast request–response cycles.
- Database transactions prevent delays during peak meal hours.

1.2.2 Scalability

- Supports increasing number of users, hostels, and bookings.
- Backend APIs can be horizontally scaled if required.

1.2.3 Maintainability

- Changes in UI do not affect backend logic.
- Business rules are centralized in controllers.
- Modular design simplifies debugging and testing.

1.2.4 Security

- Centralized authentication using JWT.
- Role-based access enforced at controller level.
- No direct database access from the client.
- Single-use, time-bound QR tokens prevent misuse.

1.2.5 Usability

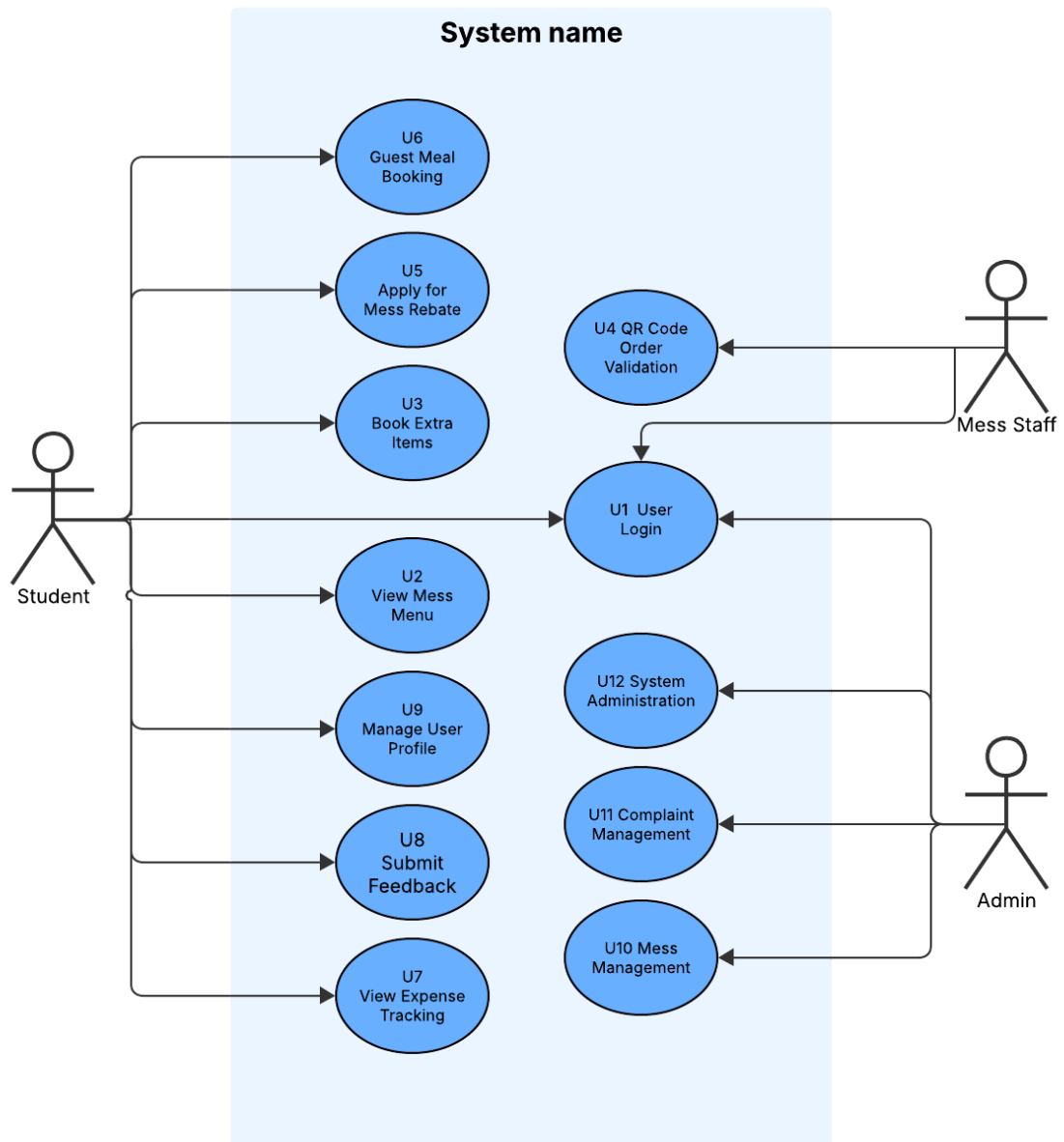
- Different views for different user roles.
- Minimal steps for common actions like booking and rebate application.
- Mobile-friendly frontend ensures accessibility.

1.2.6 Future-Readiness

- New modules (payments, analytics, nutrition data) can be added easily.
- Supports onboarding of new hostels without architectural changes.

3 Object Oriented Design

3.1 Use Case Diagrams



Registered users shall fall into three roles: **Student**, **Mess Staff**, and **Mess Administrator**. Both students and mess staff can log in to the system and access role-specific features.

Students can:

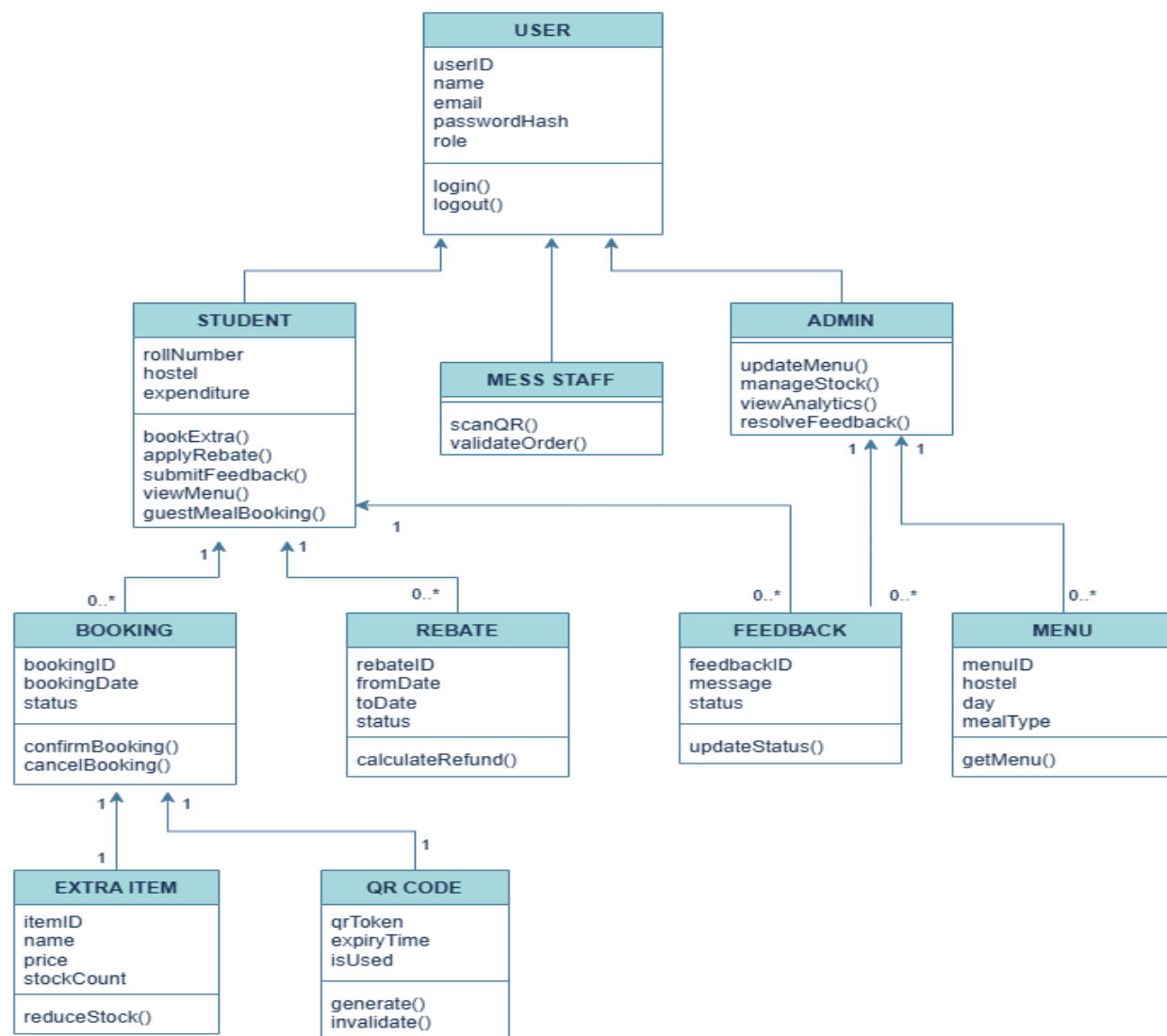
- View menus, book extra items and guest meals.

- Apply for a mess rebate and track expenses.
- Submit feedback and manage their profile.

Mess administrators, in addition to the above, can:

- Manage menus, complaints, and system configurations.

3.2 Class Diagrams



Our application follows an object oriented design pattern. The following components are represented as classes:

- Users of the system.
- Students accessing mess services.
- Mess staff responsible for order validation.

- Mess administrators managing menus and complaints.
- Bookings for extra food items.
- Extra food items available for booking.
- QR codes generated for order validation.
- Mess rebate records.
- Feedback and complaints submitted by students.
- Mess menus maintained by administrators.

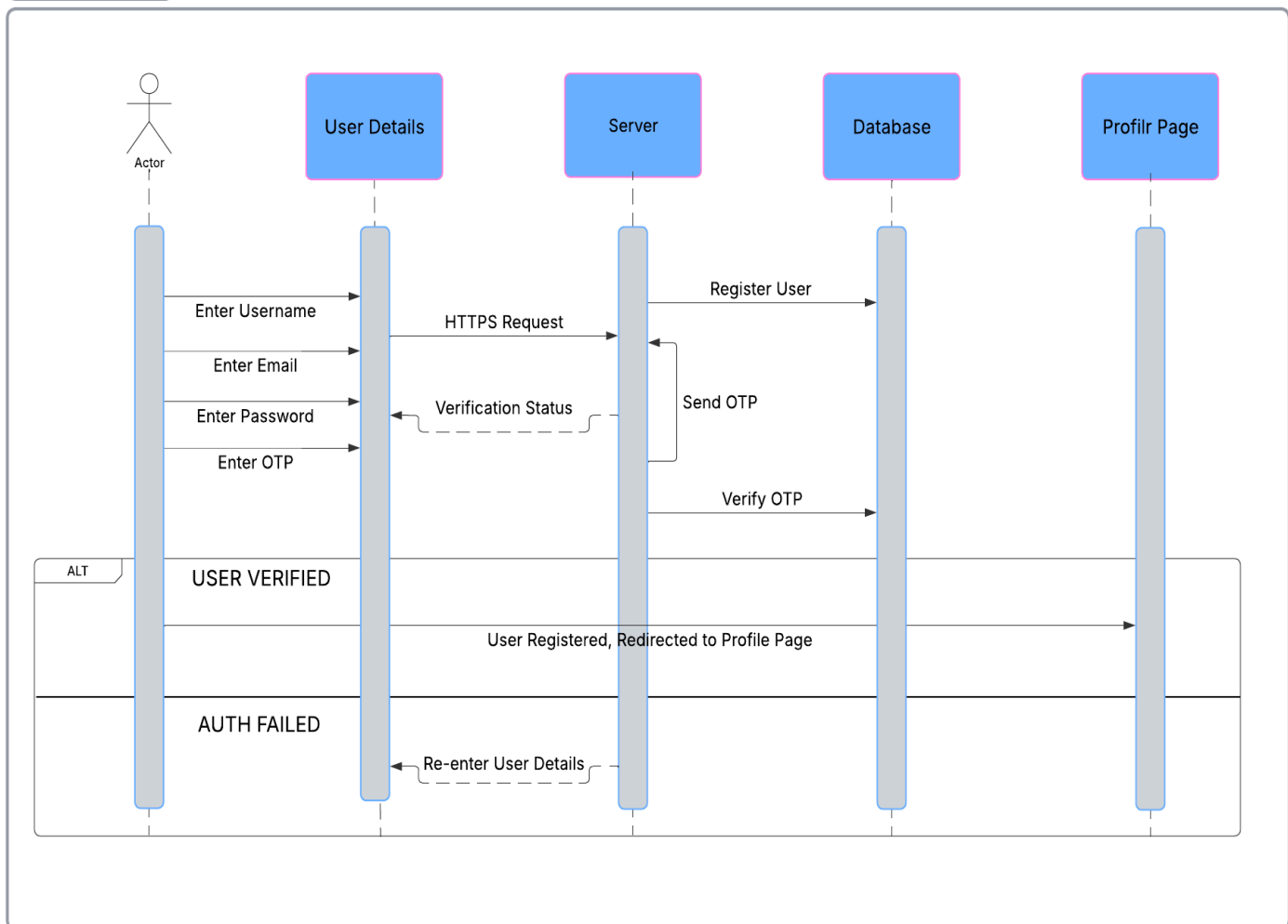
The `User` class acts as a base class and is specialized into `Student`, `MessStaff`, and `Admin` classes, enabling role-based access and functionality. Students are associated with bookings, rebates, and feedback records, while mess staff interact with bookings for QR-based validation. Administrators manage menus, inventory, and complaint resolution.

Data members of each class are encapsulated and accessed through public methods, ensuring information hiding. Class methods are responsible for modifying object state and coordinating system operations, thereby supporting a modular, maintainable, and scalable system design.

3.3 Sequence Diagrams

3.3.1 Sign Up

Sign UP

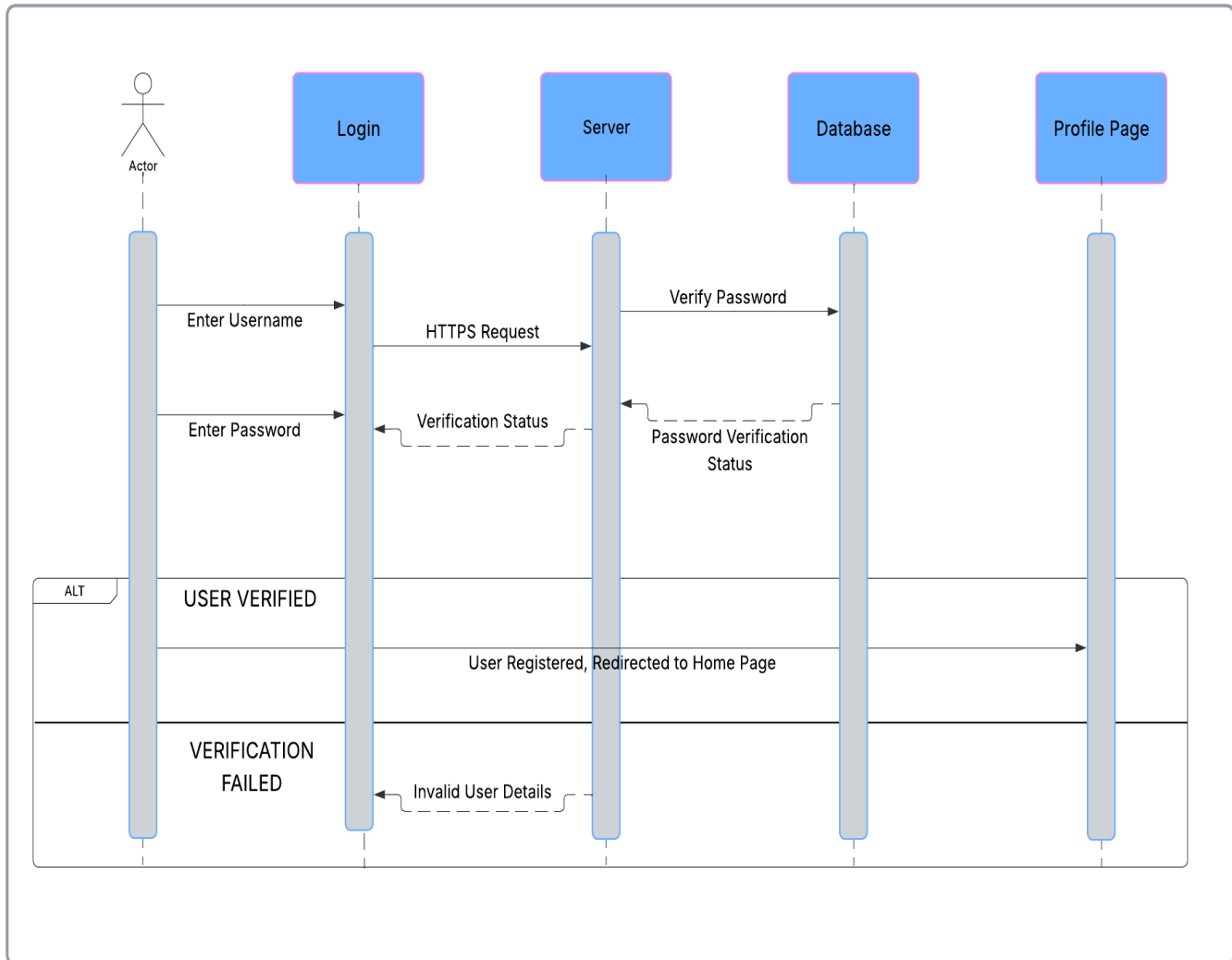


Purpose of the SignUp Page:

- To provide secure access to the website for both users and administrators.
- To verify the authenticity of all users and prevent unauthorized or malicious access.
- To assign appropriate roles (user/admin) based on permissions.
- To enforce system rules so users cannot perform actions beyond their allowed scope.
- To protect sensitive data related to mess finances and billing.
- To ensure each user is linked to their designated mess.
- To prevent misuse of meal coupons across different messes.

3.3.2 Login

Login

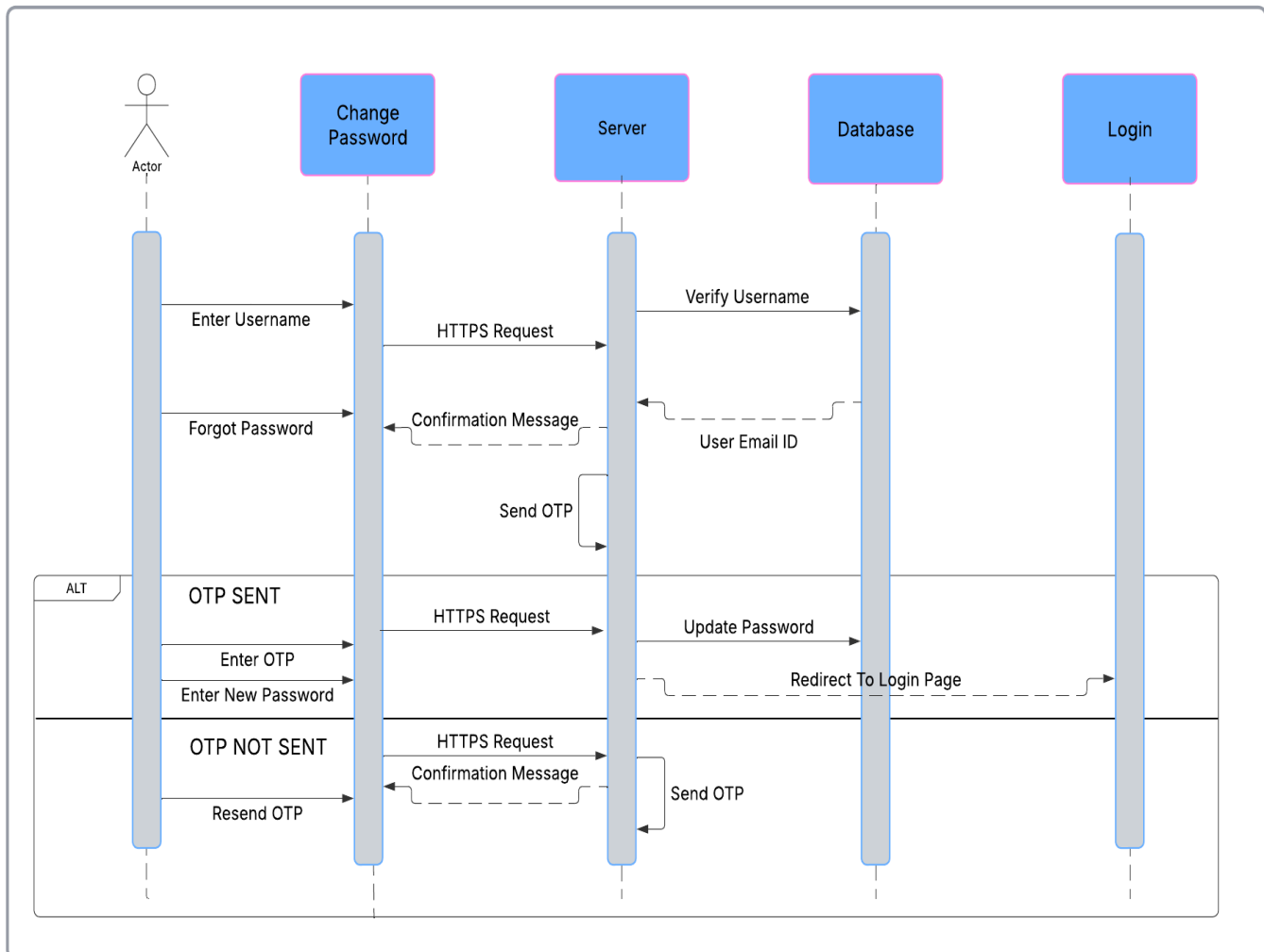


Purpose of the Login Page:

- To authenticate registered users and administrators before granting access.
- To verify login credentials and ensure only authorized users can enter the system.
- To provide role-based access after successful authentication.
- To prevent unauthorized access to sensitive mess-related data.
- To maintain security for financial records, billing, and coupon usage.
- To ensure users can only access features permitted by their assigned role.
- To protect the system from repeated invalid or malicious login attempts.

3.3.3 Change Password

Change Password

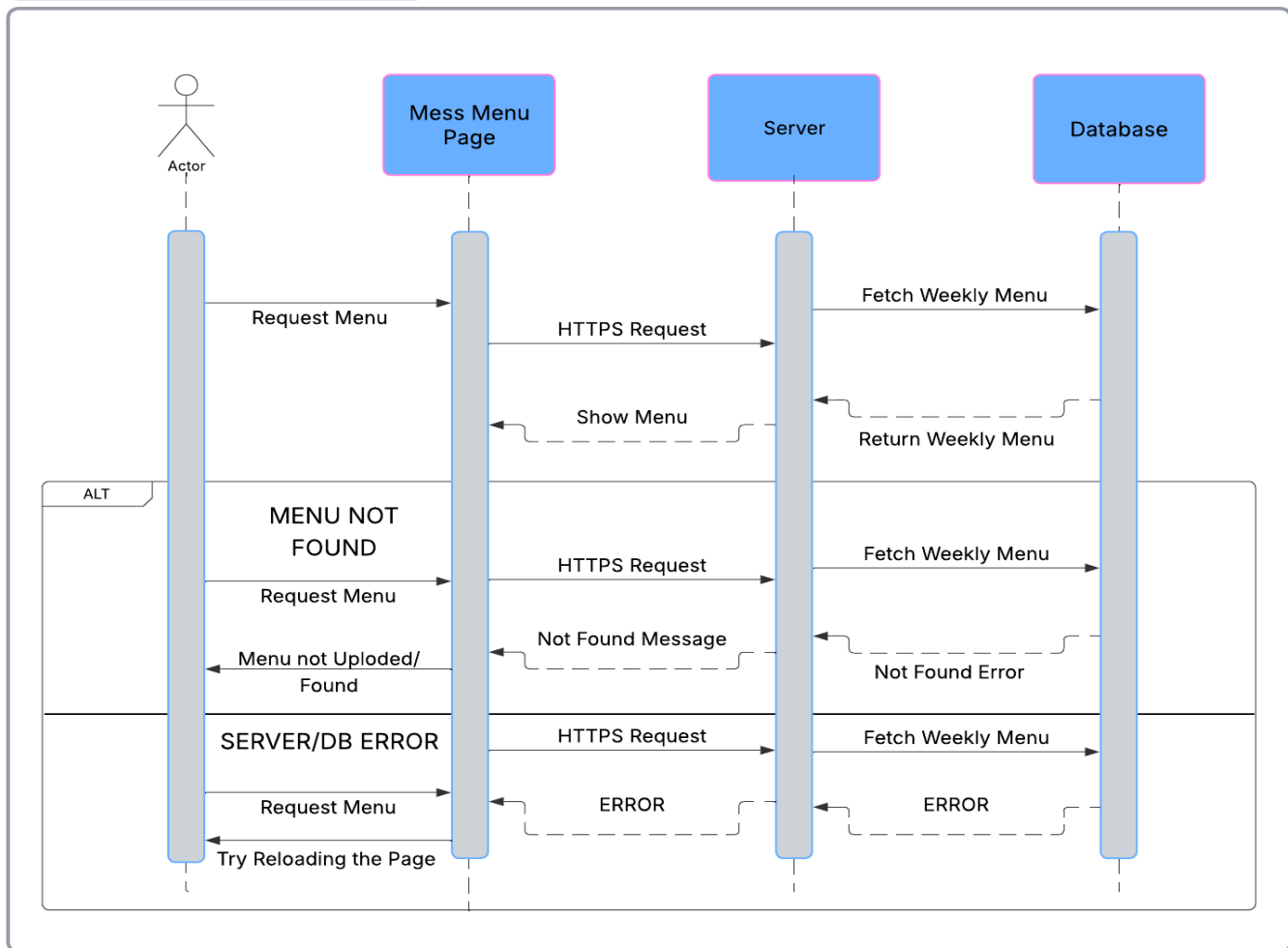


Purpose of the Forgot Password Page:

- To allow users to securely reset their password if they forget it.
- To verify the user's identity using a registered username and email.
- To ensure password reset requests are authenticated through OTP verification.
- To prevent unauthorized password changes by validating one-time credentials.
- To securely update the user's password in the database.
- To handle OTP failure cases by allowing resend functionality.
- To redirect users to the login page after successful password reset.

3.3.4 View Centralized Mess Menu

View Mess Menu

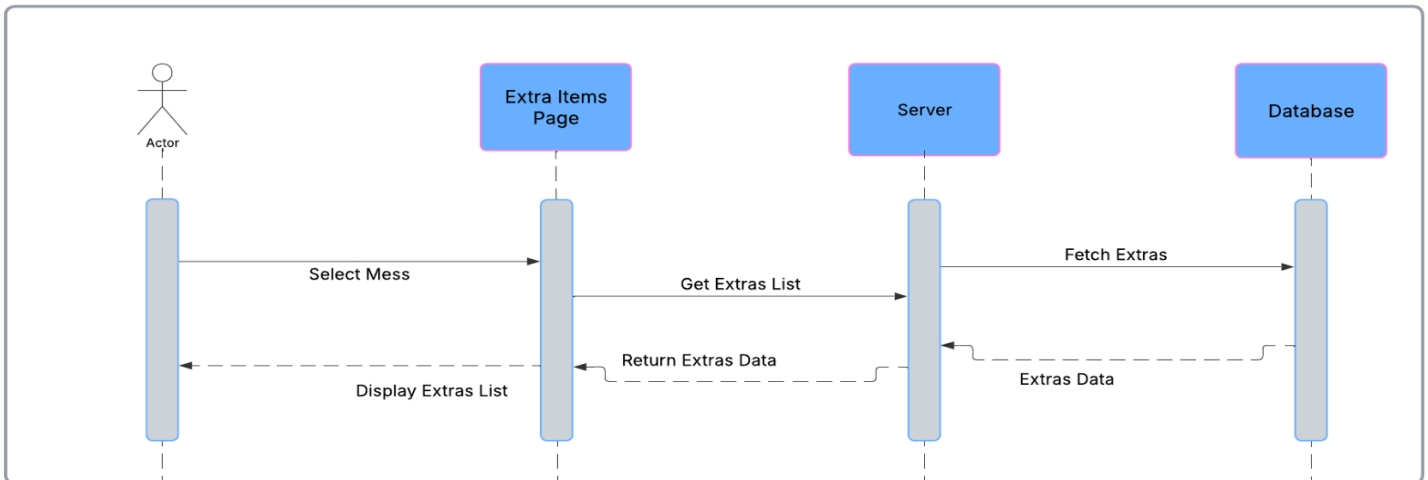


Purpose of the View Mess Menu Page:

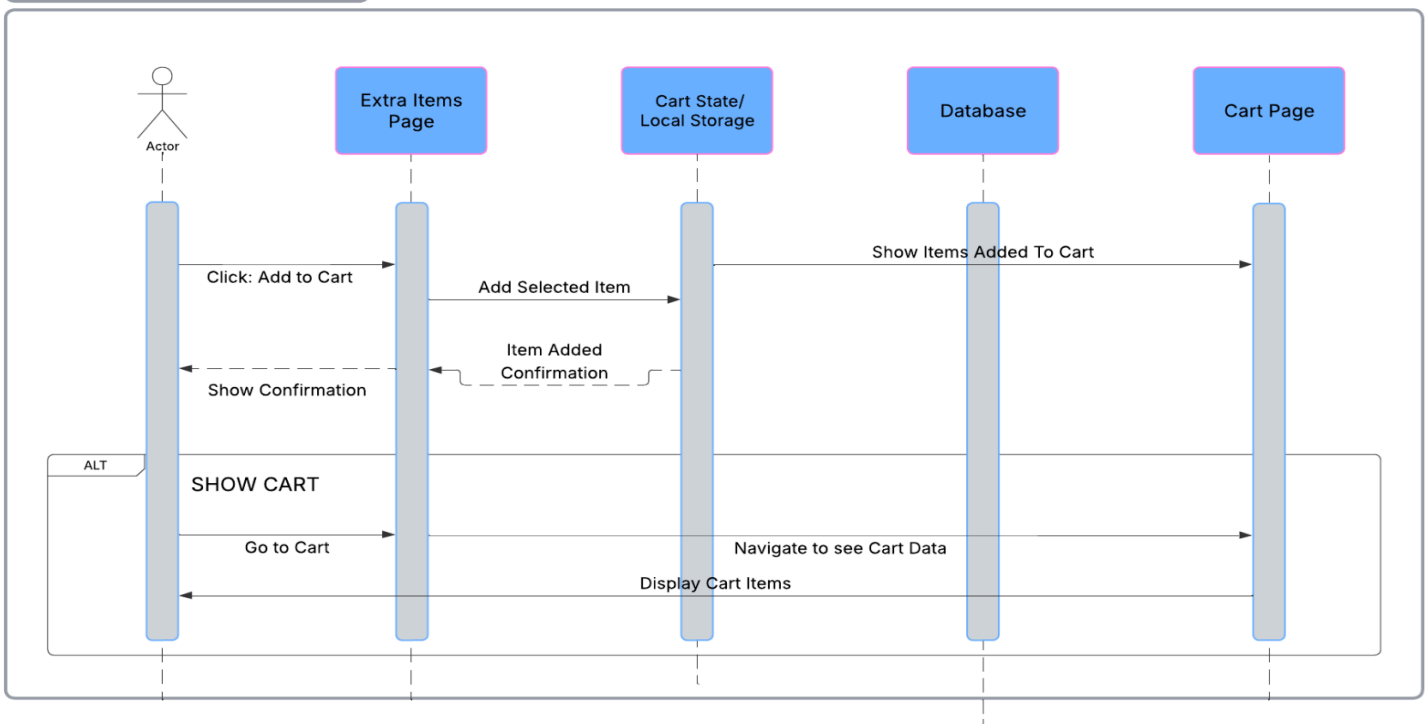
- To allow students to view the daily and weekly mess menus in a centralized manner.
- To provide accurate information about meals offered across different messes.
- To retrieve menu data securely from the server and display it in an organized format.
- To ensure students can easily navigate menus based on day and meal type.
- To handle cases where the menu is unavailable or not yet uploaded by displaying appropriate messages.
- To prevent unauthorized access to menu data by allowing only authenticated users to view menus.

3.3.5 Extra Item Booking and QR Generation

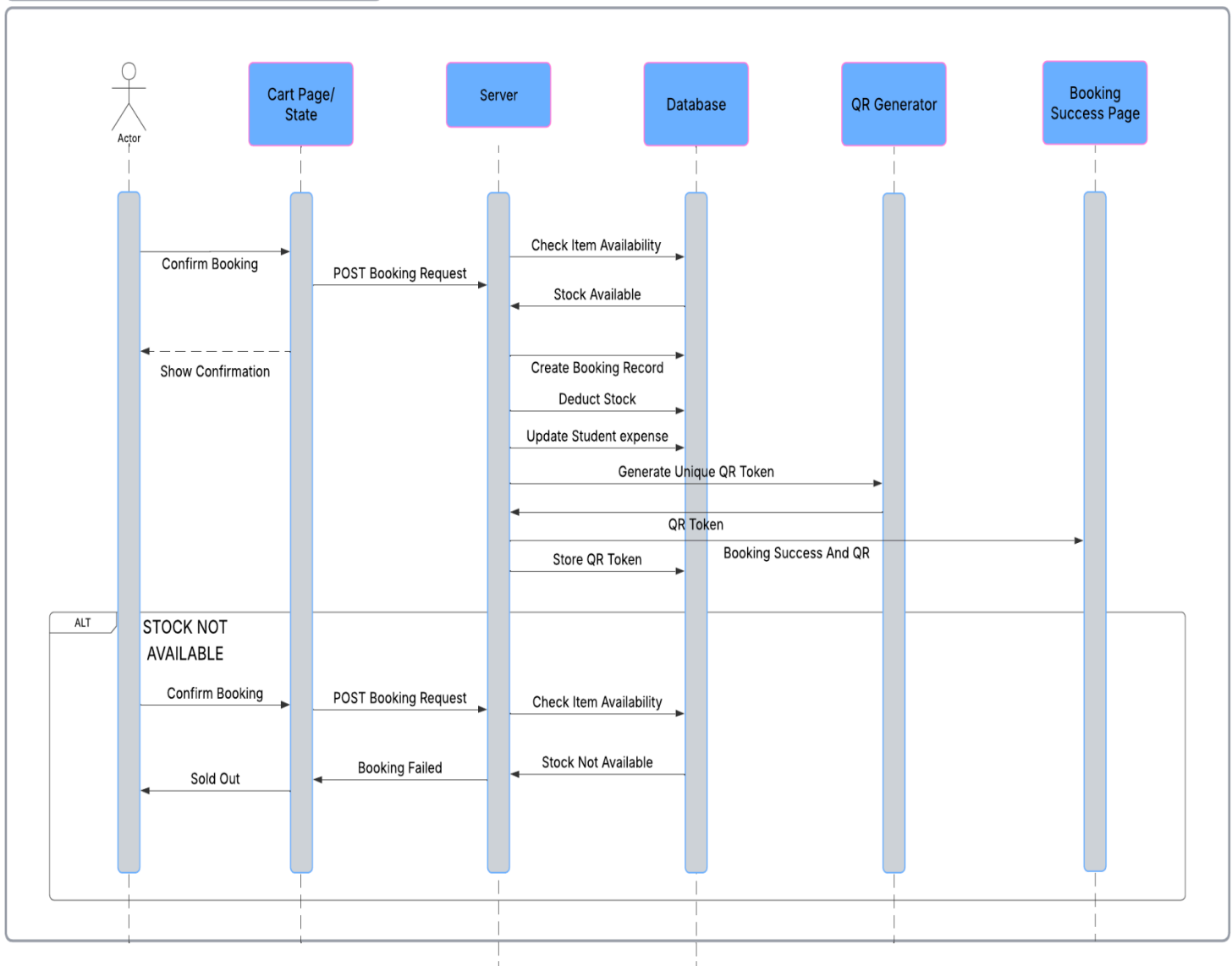
View Extra Items



Add Item To Cart



Stock Check And Booking

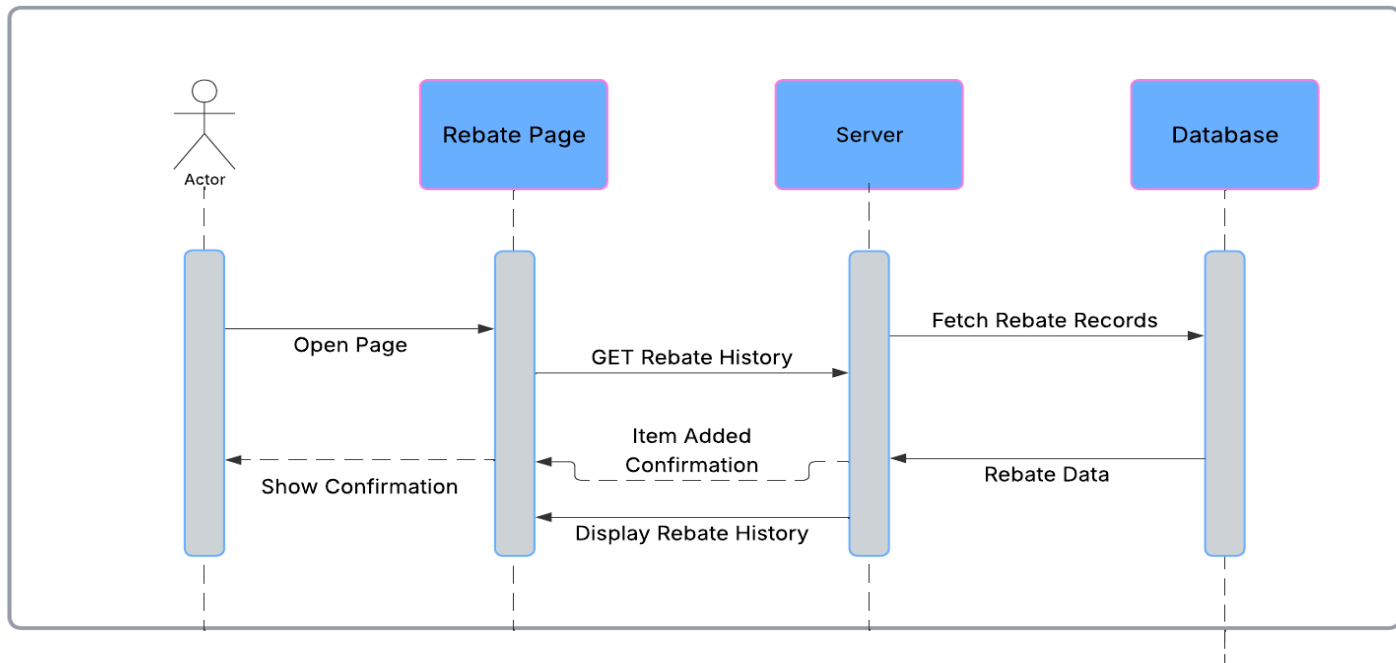


Purpose of the Booking and QR Generation:

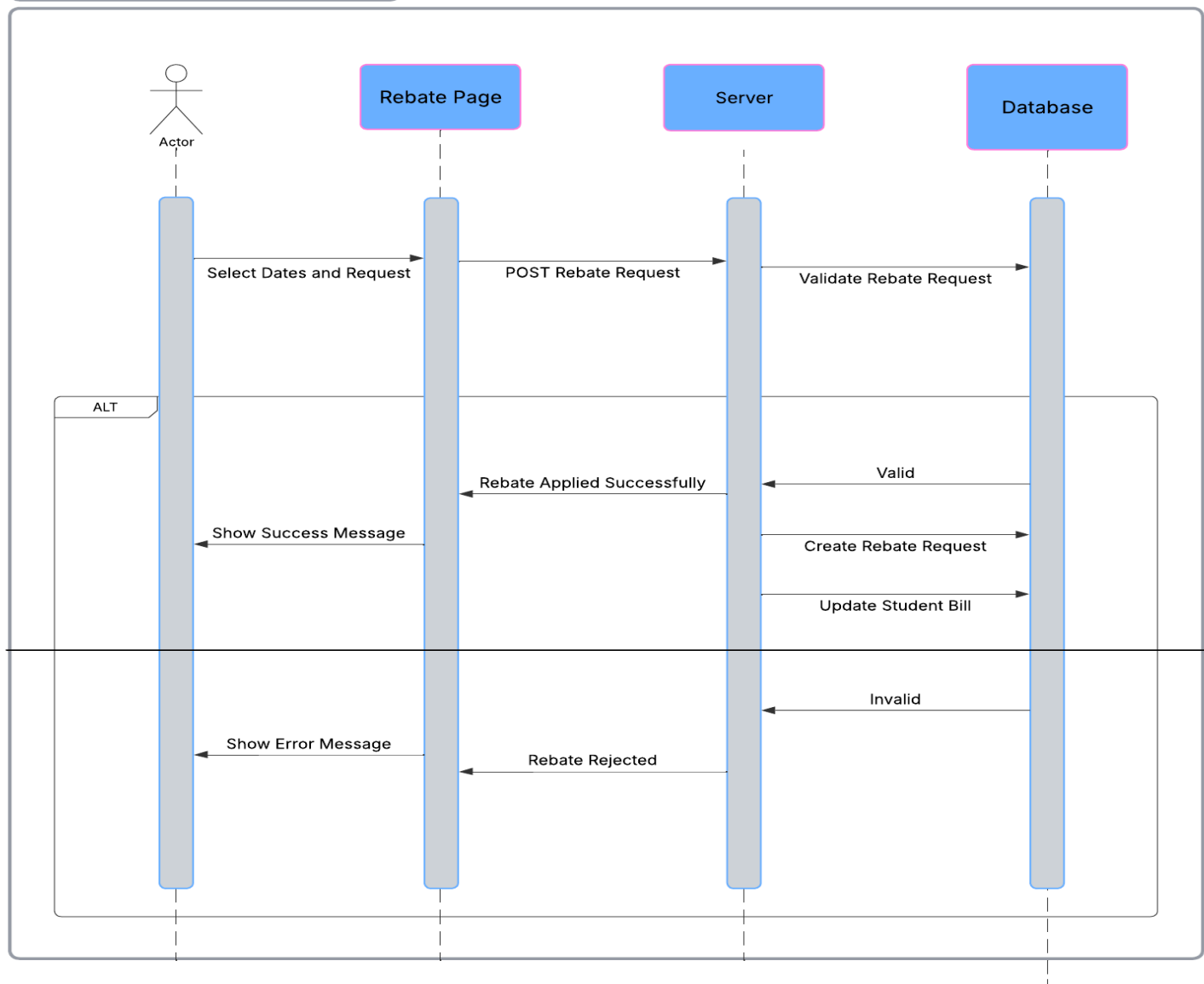
- To allow students to confirm the booking of extra items selected in the cart.
- To verify real-time availability of items before finalizing the booking.
- To generate a unique, single-use QR code for each successful booking.
- To securely store the generated QR code and associated booking details in the database.
- To update the student's expenditure automatically after a successful booking.
- To prevent overbooking and unauthorized claims through strict validation and QR-based verification.
- To redirect the user to the booking success page displaying the QR code and booking metadata.

3.3.6 Rebate Application

Rebate Page



Apply For Rebate

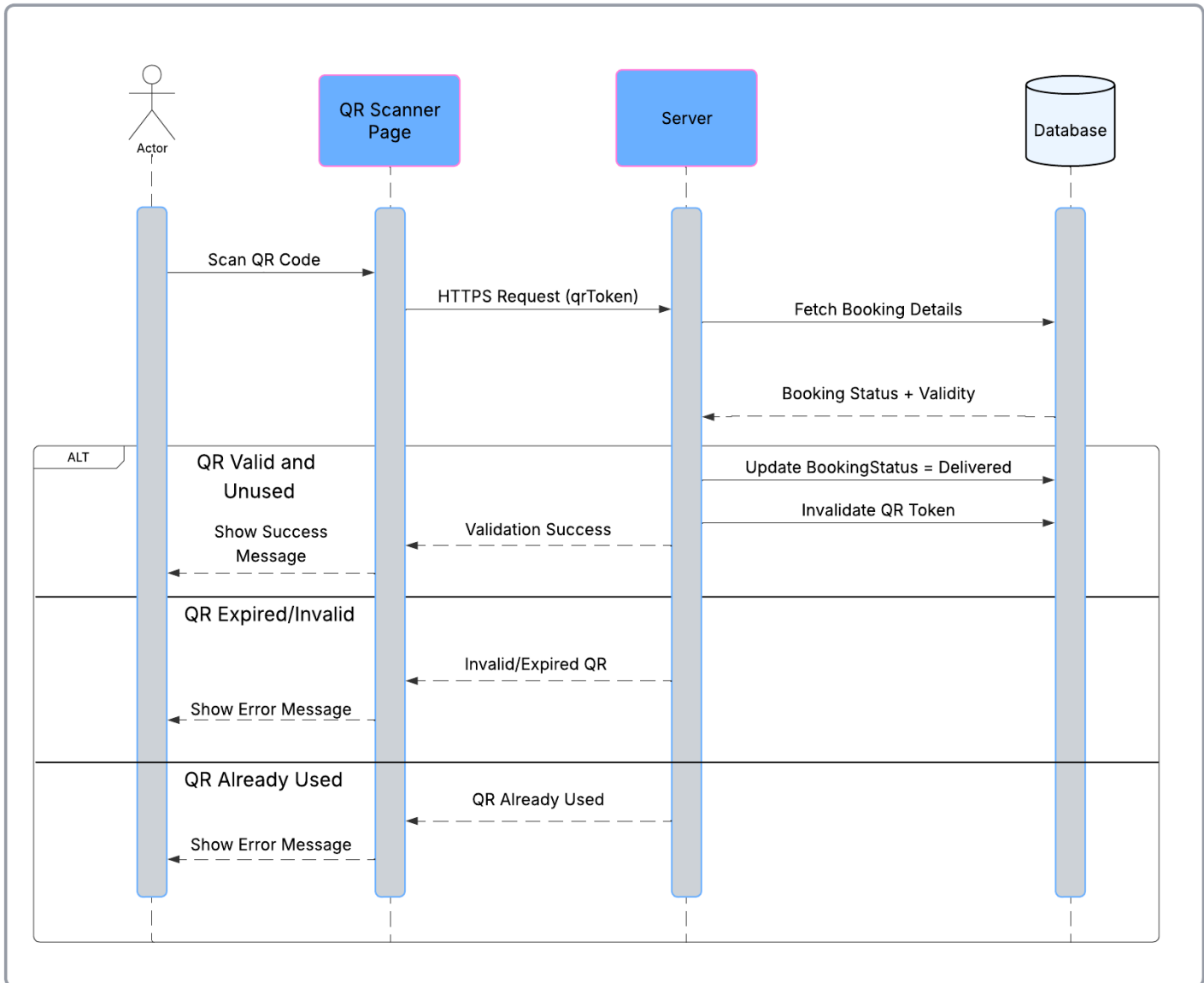


Purpose of the Booking and QR Generation:

- To allow students to apply for mess rebate (leave) for selected dates in an online and paperless manner.
- To validate rebate requests based on predefined rules such as submission deadlines and date eligibility.
- To automatically calculate the rebate amount corresponding to the approved leave period.
- To update the student's mess bill and rebate records upon successful application.
- To display the current status of rebate requests (Pending, Approved, Rejected) to the student.
- To prevent invalid or late rebate submissions through automated system checks.
- To maintain a complete and accurate history of rebate applications for transparency and auditing purposes.

3.3.7 QR-Based Order Validation

QR-Based Validation

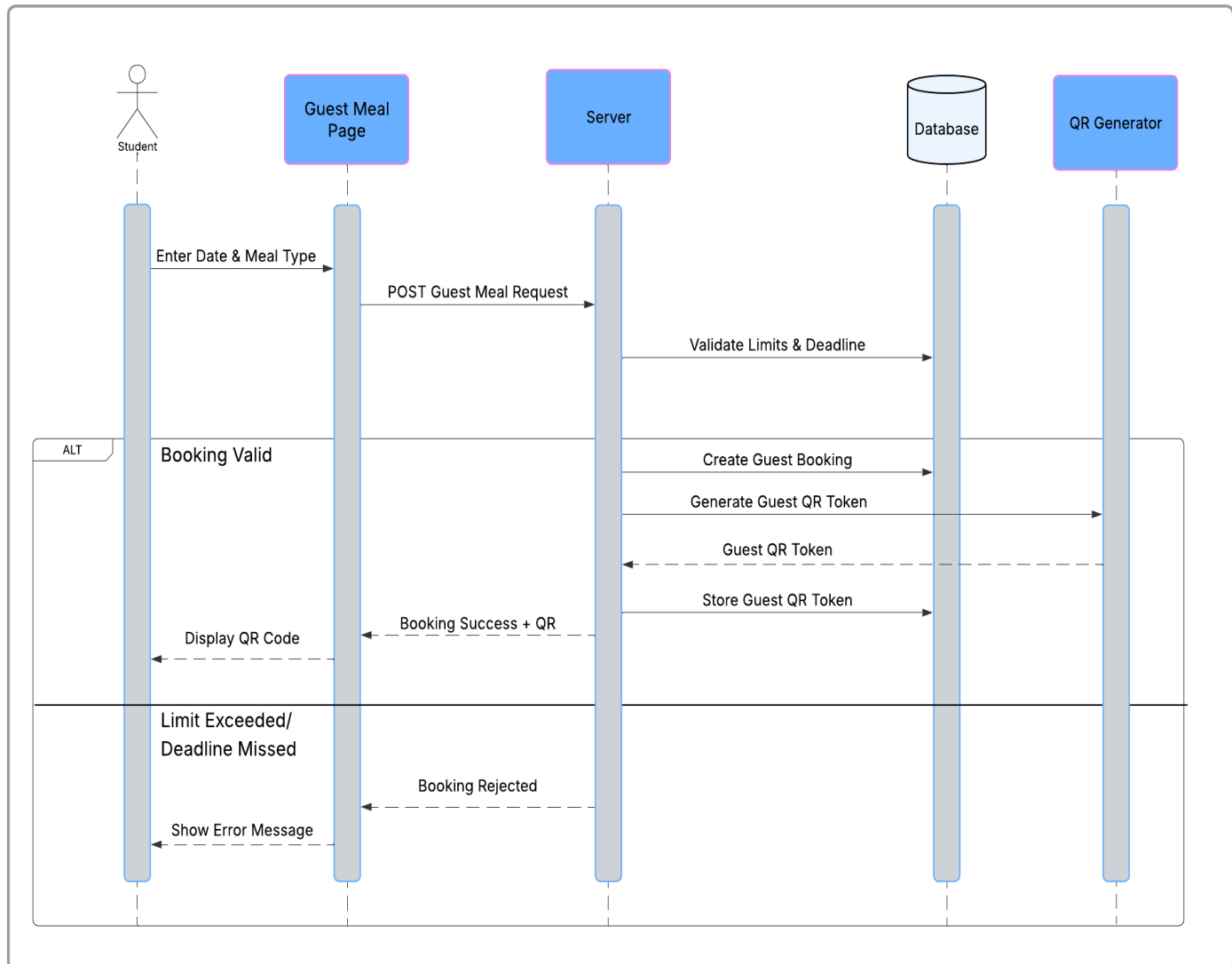


Purpose of QR-Based Order Validation

- To allow mess staff to validate student bookings using QR codes.
- To prevent reuse or forgery of booking QR codes.
- To mark bookings as *Delivered* after successful validation.
- To update inventory and booking status securely.
- To ensure fast and reliable service during peak hours.

3.3.8 Guest Meal Booking

Guest Meal Booking

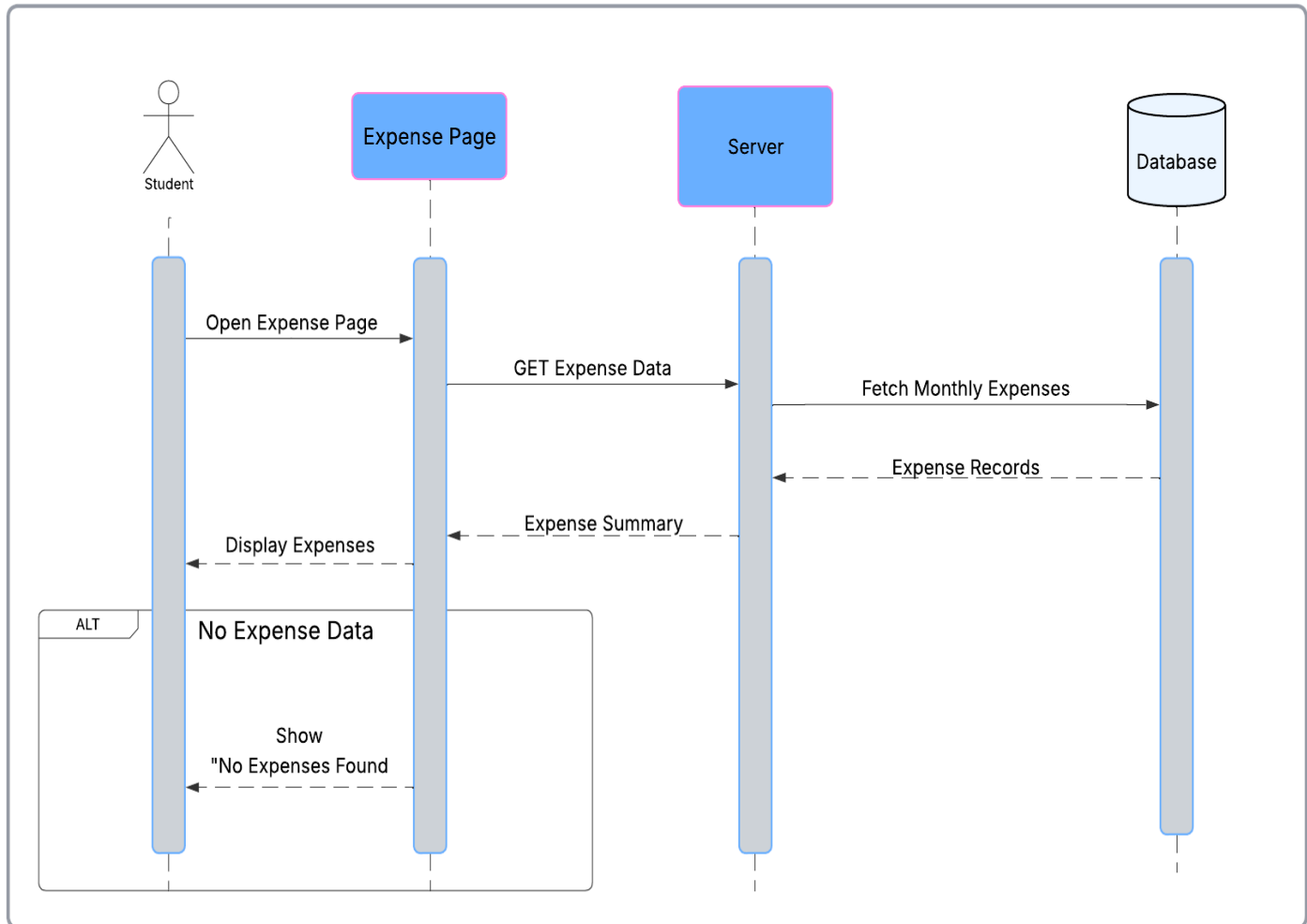


Purpose of Guest Meal Booking

- To allow students to book meals for guests digitally.
- To enforce guest meal limits and booking deadlines.
- To generate a unique guest QR token.
- To enable secure validation at the mess counter.

3.3.9 View Expense Tracking

View Expense Tracking

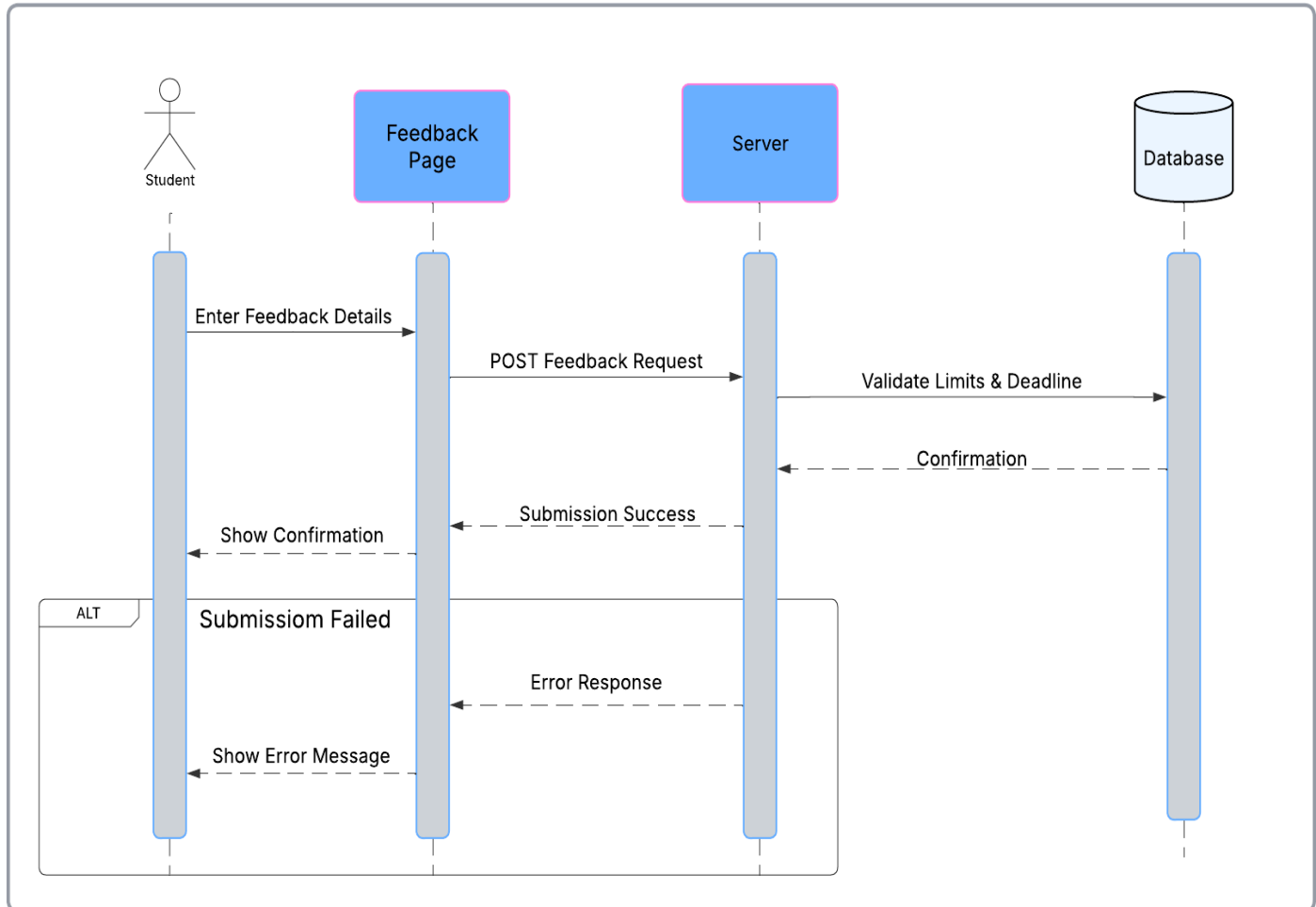


Purpose of Expense Tracking

- To allow students to view monthly expenditure.
- To ensure transparency in extra-item charges.
- To retrieve immutable expense data securely.
- To display summarized billing information.

3.3.10 Feedback / Complaint Submission

Feedback and Complaint Submission

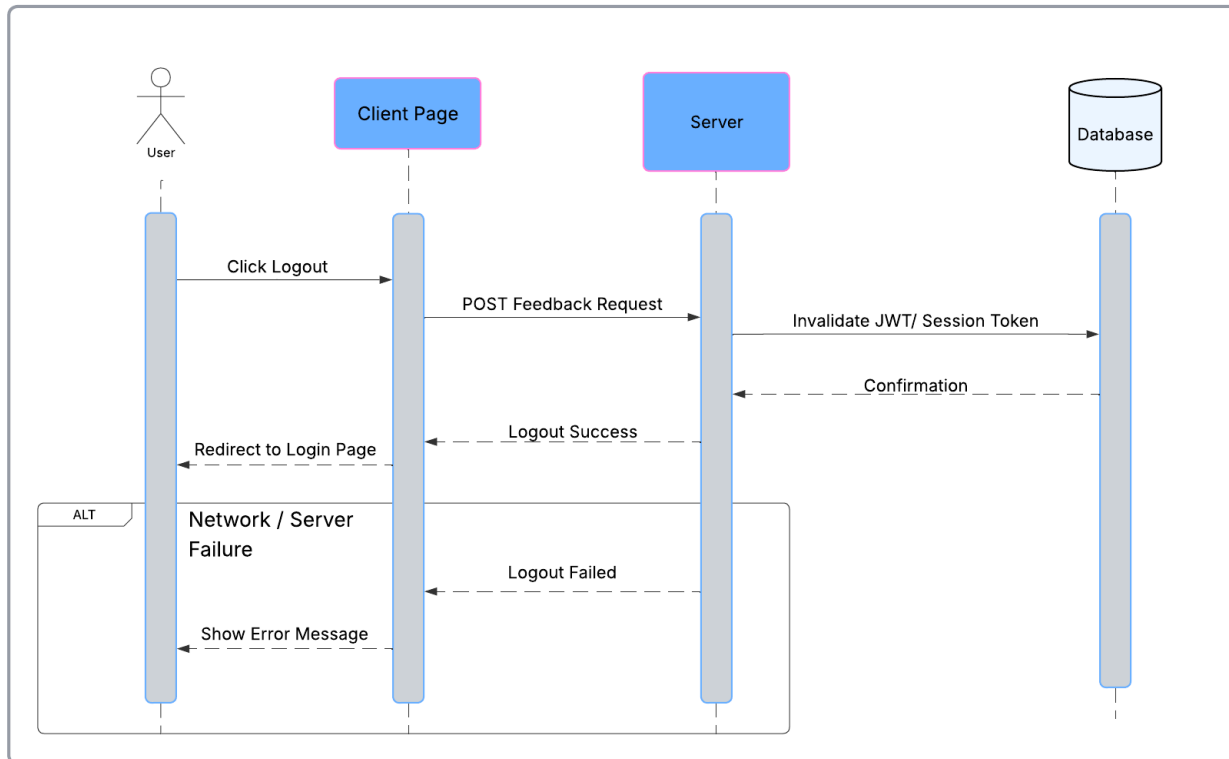


Purpose of Feedback & Complaint Submission

- To allow students to report issues digitally.
- To track complaint resolution status.
- To enable administrators to manage complaints.
- To ensure transparency and accountability.

3.3.11 User Logout

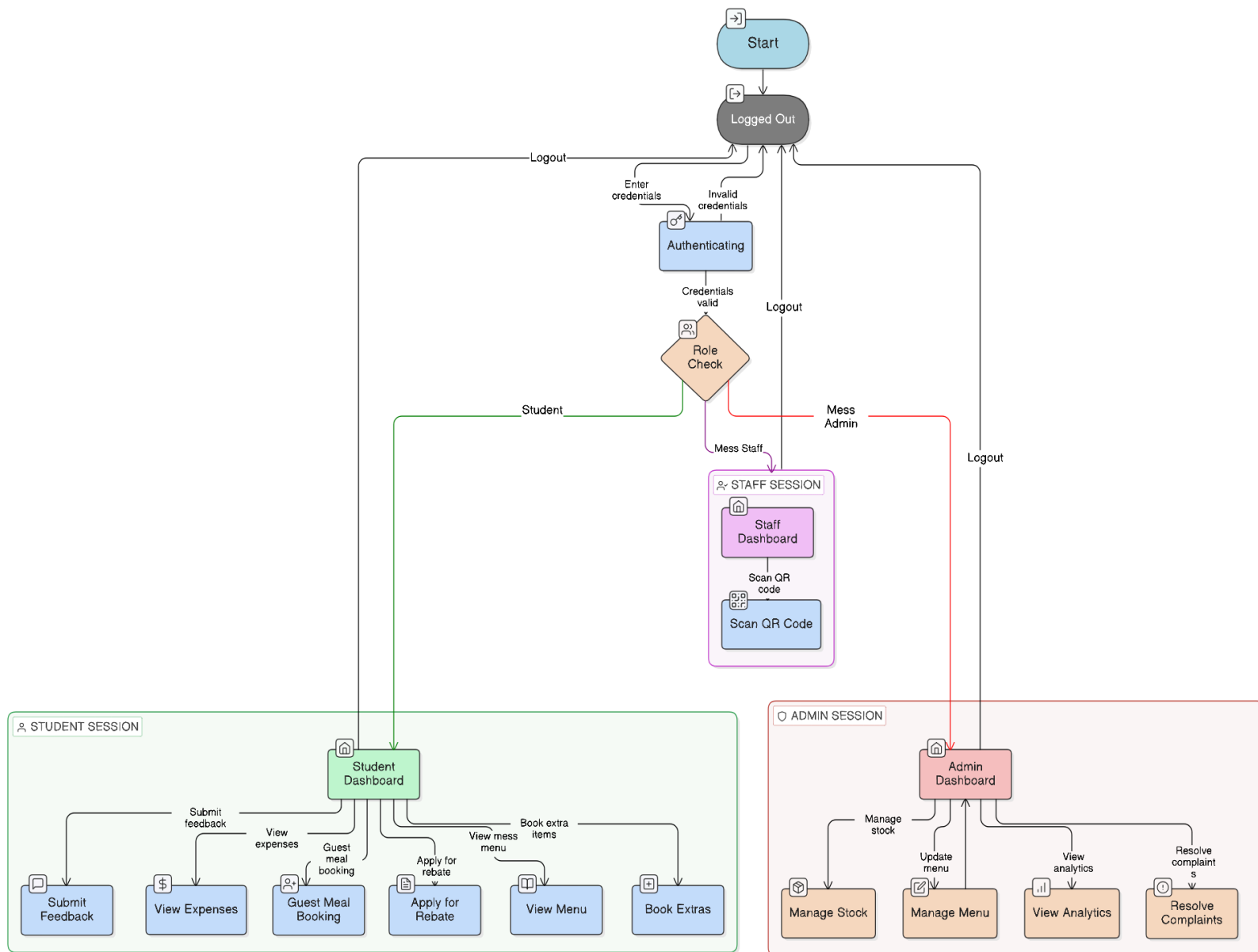
User Logout



Purpose of Logout

- To allow users to securely terminate their session.
- To invalidate authentication tokens and prevent unauthorized reuse.
- To clear client-side session data.
- To redirect users back to the login page.
- To protect sensitive mess and billing information.

3.4 State Diagrams



The state diagram illustrates the various states and transitions within the Centralized Mess Management System (CMMS). It highlights the interaction flow between different user roles—Student, Manager (Admin), and Mess Staff—as they navigate through the system's functional modules and sub-state.

3.4.1 Overview

The system architecture follows a top-down hierarchy. From the initial **LOGIN** state, the system identifies the **userRole** and transitions the user into their respective portal. Within each portal, modules are categorized into **VIEWABLE** states

(for data retrieval and tracking) and **EDITABLE** states (for transactional actions like booking, rebate applications, or status updates).

3.4.2 State Descriptions

1. Student States

- **Mess Menu (Viewable):** A passive state where the system retrieves synchronized menu data from the hall database.
- **Rebate Management (Editable):** An active state where the system validates timestamps to ensure applications are submitted at least 24 hours in advance.
- **Extras Booking (Editable):** A transactional state involving real-time stock verification and unique QR token generation.

2. Manager (Admin) States

- **Analytics (Viewable):** Aggregates data from approved rebates and active bookings to provide real-time food requirement forecasts.
- **Grievance Management (Editable):** Allows the admin to transition a complaint state from "Open" to "In Progress" or "Resolved."

3. Mess Staff States

- **Scanner Module (Editable):** A critical security state that decodes student QR tokens, validates single-use integrity, and finalizes the "Delivered" state for the order.

4 Project Plan

The development of the Centralized Mess Management System (CMMS) adheres to a strict academic schedule, ensuring all milestones from requirement analysis to final presentation are met efficiently.

4.1 Communication and Collaboration

To ensure seamless coordination among the members of *FullThrottle Tenacious Dynasty*, the following protocols have been established:

- **Communication:** Daily updates via WhatsApp; technical documentation on Discord.
- **Version Control:** GitHub repository with feature-branch workflows for React (Frontend) and Django (Backend).

4.2 Project Timeline

The project timeline is divided into specific weeks with strict deadlines and defined deliverables as detailed below.

TimeFrame	Deadline	Task	Deliverable
Week 1	Jan 7	Team registration	—
Week 1	Jan 11	Project registration	—
Week 2–3	Jan 23	Requirement generation, analysis, and documentation	Requirement Document
Week 4–5	Feb 6	Object-oriented design	Design Document
Week 6–10	Mar 27	Implementation, unit testing, integration testing	Implementation Document, Code in GitHub Repository
Week 11	Apr 3	System testing, manual preparation for beta testing	Test Document, System Manual
Week 12	Apr 12	Code improvement, beta testing of some other project, addressing beta testing feedback	Beta Test Report
Week 13-14	Apr 24	Project Presentations	Final Project Report

4.3 Gantt Chart

The visual representation of the project timeline is provided below.

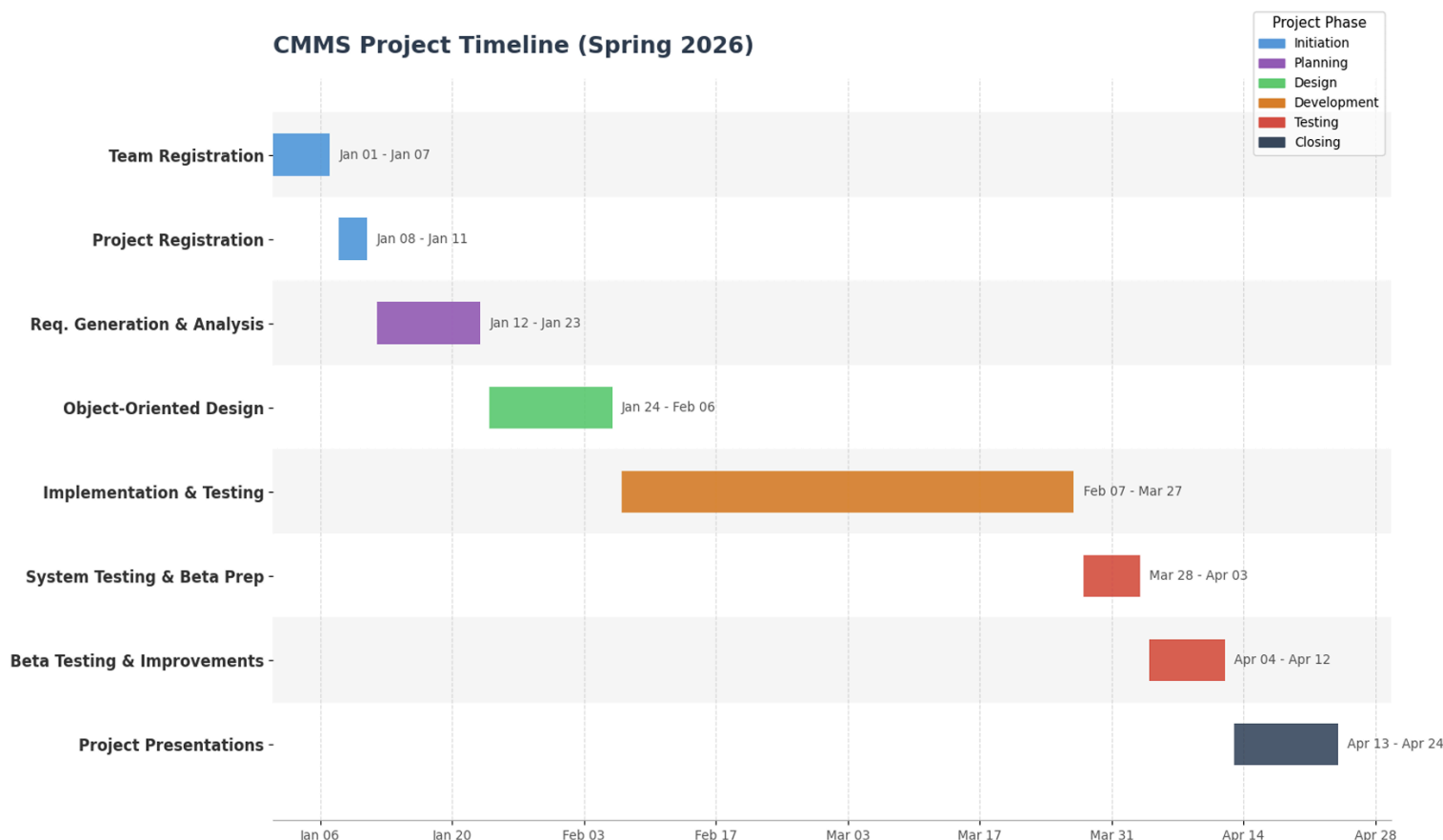


Figure 1: Project Phases and Milestones (2026)

4.4 Development Phases and Team Responsibilities

The development tasks are divided based on the technology stack defined in the SRS (React.js for Frontend and Django for Backend).

4.4.1 Frontend Development

Responsibilities: Implementation of the User Interface using Vite + React.js, styling with Tailwind CSS, and animations via Framer Motion.

Frontend Developer Team

- Shital Niras
- Kavita Kumar
- Deepak Kumar
- Ashwina Yadav
- Itesh Mishra
- Tanush Khagwal

4.4.2 Backend Development Team

Responsibilities: Development of the REST API using Django and Django REST Framework, database management (PostgreSQL), and authentication (JWT).

Backend Developer Team

- Aaditya Bijarniya
- Shubham Kumar Patel
- Dev Prakash
- Soumyaranjan Sahu
- Deepak Kumar
- Tanush Khagwal

5 Other Details

Term	Definition
Student	Any student accessing the website.
Mess Staff	Any Staff accessing the mess staff.
Admin	Administrator of the website.
Guest	Any person other than the user. For whom meal is being booked from user's account
Rebate	Leave from Mess.
Authentication	Process of verifying user identity
Extra Items	Food Items not included in the main menu.

Appendix A - Group Log

Date & Time	Venue	Activity
24/01/2026	1st floor, Rajeev Motwani Building	Discussed the architecture of the website and its implementation.
30/01/2026	1st floor, Rajeev Motwani Building	Distributed the work among teammates.
01/02/2026	Online Meet	Reviewed the work progress and suggested improvements in it.
06/02/2026	Online Meet	Final review of the SDS document.

